# 如何启用CobaltStrike隐藏的Sleep_mask功能

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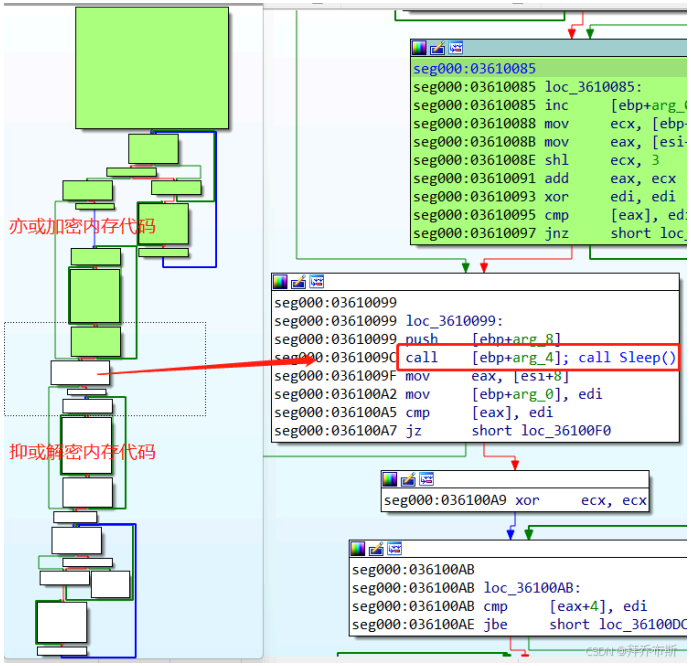
## Sleep_mask()功能作用

CobaltStrike信标运行期间会执行Sleep()函数，为了防止进程被检测会在Sleep()之前使用亦或算法加密内存中代码，Sleep()结束之后再使用亦或算法解密出内存中代码；CobaltStrike信标大部分时间处于Sleep()中，进程中代码加密后就会难于检测。Sleep_mask函数的缩略图如下图所示：




即使CobaltStrike信标的内存代码被加密，依旧可以通过检测内存中的Sleep_mask函数来检测CobaltStrike信标的存在：

```
C:\Users\Administrator>D:\Desktop\yara64.exe D:\Desktop\sleep_mask.yar 1292
CobaltStrike_Sleep_Mask_32 1292

C:\Users\Administrator>
```

CSDN @拜乔布斯

# CobaltStrike逆向学习系列(6)：Beacon sleep_mask 分析

安全工具 (https://forum.butian.net/topic/53)

只有对工具足够的了解，才能够更好的进行修改

这是[**信安成长计划**]的第 6 篇文章

CobaltStrike 提供了一个内存混淆功能，它会在 Sleep 的时候将自身混淆从而避免一定的检测

## 0x01 C2Profile 分析

因为 sleep_mask 是从 C2Profile 中设置的，所以就需要先搞清楚 TeamServer 是如何解析的

```
setupPivotFrames(this.c2profile, settings);
this.setupKillDate(settings);
//处理 sleep_mask
this.setupGargle(settings, beaconPath);
(new ProcessInject(this.c2profile)).apply(settings);
byte[] var24 = settings.toPatch();
```

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很明显它还跟其他的设置项有关，这里我们重点关注一下 rwx 的设置

```
protected void setupGargle(Settings settings, String beaconPath) throws IOException {
    if (!this.c2profile.option( s: ".stage.sleep_mask")) {
        settings.addInt(41, 0);
    } else {
        PEParser beaconParser = PEParser.load(SleevedResource.readResource(beaconPath));
        boolean var4 = this.c2profile.option( s: ".stage.obfuscate");
        boolean var5 = this.c2profile.option( s: ".stage.userwx");
        int var6 = beaconParser.sectionEnd( s: ".text");
        settings.addInt(41, var6);
        int var7 = beaconParser.sectionAddress( s: ".rdata") - var6;
        if (var7 < 256) {
            CommonUtils.print_error( s: ".stage.sleep_mask is true; nook spa
        }
```

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首先会将 text 段的结尾地址存储到 index=41 的位置

```
    boolean var4 = this.c2profile.option( s: ".stage.obfuscate");
    boolean var5 = this.c2profile.option( s: ".stage.userwx");
    int var6 = beaconParser.sectionEnd( s: ".text");
    settings.addInt(41, var6);
    int var7 = beaconParser.sectionAddress( s: ".rdata") - var6;
    if (var7 < 256) {
```

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接着判断了 text 段与 rdata 段中间的空白位置够不够 256 个字节，推测会将加解密函数放在这里

```
int var6 = beaconParser.sectionEnd( s: ".text");
settings.addInt(41, var6);
int var7 = beaconParser.sectionAddress( s: ".rdata") - var6;
if (var7 < 256) {
    CommonUtils.print_error( s: ".stage.sleep_mask is true; nook space in " + beaconPath + " is " + var7 + " by
}
```

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obfuscate 就让它默认为 false，它不是我们这次关注的重点，接下来就会将 0 和 4096 添加到其中

```
Packer packer = new Packer();
packer.little();
if (!obfuscate) {
    packer.addInt( i: 0);
    packer.addInt( i: 4096);
}
```

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再看一眼 text 段的 Virtual Address，这块很明显就是 PE 头的那段内容

| Name | Virtual Size | Virtual Address | Raw Size | Raw Address | Reloc Ad |
|---|---|---|---|---|---|
| 00000200 | 00000208 | 0000020C | 00000210 | 00000214 | 00000218 |
| Byte[8] | Dword | Dword | Dword | Dword | Dword |
| .text | 0002ABF2 | 00001000 | 0002AC00 | 00000400 | 00000000 |
| .rdata | 0000F8B2 | 0002C000 | 0000FA00 | 0002B000 | 00000000 |
| .data | 0000D268 | 0003C000 | 00002600 | 0003AA00 | 00000000 |

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接下来就是循环添加内容了，可以很明显的看到 do while 的循环条件是 text 段且不允许使用 rwx，这也就意味着，当我们不允许使用 rwx 的时候，text 段并不会被添加到 index=42 的项中，应该也就不会被混淆了，在最后还添加了零，用于标示结尾

```
Iterator sectionsTable = beaconParser.SectionsTable().iterator();

while(true) {
    String var10;
    do {
        if (!sectionsTable.hasNext()) {
            packer.addInt( i: 0);
            packer.addInt( i: 0);
            settings.addData(42, packer.getBytes(), (int)packer.size());
            return;
        }

        var10 = (String)sectionsTable.next();
    } while(".text".equals(var10) && !userwx);

    packer.addInt(beaconParser.sectionAddress(var10));
    packer.addInt(beaconParser.sectionEnd(var10));
}
```

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# 0x02 set userwx "true"

为了快速定位到加解密函数所在，可以设置断点来完成，因为按照之前的分析，使用 rwx 的时候，text 段也会被混淆的，所以跟入函数以后，直接滑倒开头下写入断点即可

```
0000000000920000   4D:5A                 pop r10
0000000000920002   41:52                 push r10
0000000000920004   55                    push rbp
0000000000920005   48:89E5               mov rbp,rsp
0000000000920008   48:81EC 20000000      sub rsp,20
000000000092000F   48:8D1D EAFFFFFF      lea rbx,qword ptr ds:[920000]      rbx:L"褂 圈 [illegible]"
0000000000920016   48:89DF               mov rdi,rbx                        rbx:L"褂 圈 [illegible]"
0000000000920019   48:81C3 F45F0100      add rbx,15FF4                      rbx:L"褂 圈 [illegible]"
0000000000920020   FFD3                  call rbx
0000000000920022   41:B8 F0B5A256        mov r8d,56A2B5F0
0000000000920028   68 04000000           push 4
000000000092002D   5A                    pop rdx
000000000092002E   48:89F9               mov rcx,rdi
0000000000920031   CC                    int3
0000000000920032   D000                  rol byte ptr ds:[rax],1
0000000000920034   0000                  add byte ptr ds:[rax],al
0000000000920036   0000                  add byte ptr ds:[rax],al
0000000000920038   0000                  add byte ptr ds:[rax],al
000000000092003A   0000                  add byte ptr ds:[rax],al
000000000092003C   F8                    clc
000000000092003D   0000                  add byte ptr ds:[rax],al
000000000092003F   000E                  add byte ptr ds:[rsi],cl
0000000000920041   1F                    ???
0000000000920042   BA 0E00B409           mov edx,9B4000E
0000000000920047   CD 21                 int 21
0000000000920049   B8 014CCD21           mov eax,21CD4C01
000000000092004E   54                    push rsp
000000000092004F   68 69732070           push 70207369
0000000000920054   72 6F                 jb 9200C5
0000000000920056   67:72 61              jb 9200BA
0000000000920059   6D                    insd
000000000092005A   2063 61               and byte ptr ds:[rbx+61],ah
000000000092005D   6E                    outsb
000000000092005E   6E                    outsb
000000000092005F   6F                    outsd
0000000000920060   74 20                 je 920082
0000000000920062   62                    ???
0000000000920063   65:2072 75            and byte ptr gs:[rdx+75],dh
0000000000920067   6E                    outsb
```

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接着直接放过也就到达了位置

```
00000000001BBC3B   41:8BC1                mov eax,r9d
00000000001BBC3E   C1EA 02                shr edx,2
00000000001BBC41   41:FFC1                inc r9d
00000000001BBC44   6BD2 0D                imul edx,edx,D
00000000001BBC47   2BC2                   sub eax,edx
00000000001BBC49   8A4C18 10              mov cl,byte ptr ds:[rax+rbx+10]
00000000001BBC4D   41:300C38              xor byte ptr ds:[r8+rdi],cl
00000000001BBC51   48:FFC7                inc rdi
00000000001BBC54   45:3BCB                cmp r9d,r11d
00000000001BBC57   72 D7                  jb 1BBC30
00000000001BBC59   EB B8                  jmp 1BBC13
00000000001BBC5B   8BCE                   mov ecx,esi
00000000001BBC5D   FFD5                   call rbp
00000000001BBC5F   4C:8B53 08             mov r10,qword ptr ds:[rbx+8]
00000000001BBC63   45:8B0A                mov r9d,dword ptr ds:[r10]
00000000001BBC66   45:8B5A 04             mov r11d,dword ptr ds:[r10+4]
00000000001BBC6A   4D:8D52 08             lea r10,qword ptr ds:[r10+8]
00000000001BBC6E   45:85C9                test r9d,r9d
00000000001BBC71   75 05                  jne 1BBC78
00000000001BBC73   45:85DB                test r11d,r11d
00000000001BBC76   74 33                  je 1BBCAB
00000000001BBC78   45:3BCB                cmp r9d,r11d
00000000001BBC7B   73 E6                  jae 1BBC63
00000000001BBC7D   49:8BF9                mov rdi,r9
00000000001BBC80   4C:8B03                mov r8,qword ptr ds:[rbx]
00000000001BBC83   B8 4FECC44E            mov eax,4EC4EC4F
00000000001BBC88   41:F7E1                mul r9d
00000000001BBC8B   41:8BC1                mov eax,r9d
00000000001BBC8E   C1EA 02                shr edx,2
00000000001BBC91   41:FFC1                inc r9d
00000000001BBC94   6BD2 0D                imul edx,edx,D
00000000001BBC97   2BC2                   sub eax,edx
00000000001BBC99   8A4C18 10              mov cl,byte ptr ds:[rax+rbx+10]
00000000001BBC9D   41:300C38              xor byte ptr ds:[r8+rdi],cl
00000000001BBCA1   48:FFC7                inc rdi
00000000001BBCA4   45:3BCB                cmp r9d,r11d
00000000001BBCA7   72 D7                  jb 1BBC80
00000000001BBCA9   EB B8                  jmp 1BBC63
00000000001BBCAB   48:8B5C24 30           mov rbx,qword ptr ss:[rsp+30]
```

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接着根据这个位置到 IDA 中进行查找，然后通过回溯也很容易能够找到整个的调用链

```
EncAndDec       proc near               ; DATA XREF: sub_180010B18↓o
                                        ; .pdata:000000018004ABB8↓o

arg_0           = qword ptr  8
arg_8           = qword ptr  10h
arg_10          = qword ptr  18h

                mov     [rsp+arg_0], rbx
                mov     [rsp+arg_8], rbp
                mov     [rsp+arg_10], rsi
                push    rdi
                sub     rsp, 20h
                mov     r10, [rcx+8]
                mov     esi, r8d
                mov     rbp, rdx
                mov     rbx, rcx

loc_180010A69:                          ; CODE XREF: EncAndDec+39↓j
                                        ; EncAndDec+67↓j
                mov     r9d, [r10]
                mov     r11d, [r10+4]
                lea     r10, [r10+8]
                test    r9d, r9d
                jnz     short loc_180010A7E
                test    r11d, r11d
                jz      short loc_180010AB1

loc_180010A7E:                          ; CODE XREF: EncAndDec+2F↑j
                cmp     r9d, r11d
                jnb     short loc_180010A69
                mov     rdi, r9

loc_180010A86:                          ; CODE XREF: EncAndDec+65↓j
                mov     r8, [rbx]
                mov     eax, 4EC4EC4Fh
                mul     r9d
                mov     eax, r9d
                shr     edx, 2
                inc     r9d
                imul    edx, 0Dh
                sub     eax, edx
                mov     cl, [rax+rbx+10h]
                xor     [r8+rdi], cl
                inc     rdi
                cmp     r9d, r11d
                jb      short loc_180010A86
                jmp     short loc_180010A69
```

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在主循环的最后有一个用于处理 Sleep 的函数

```
      sub_180019104();
    if ( !SleepTime )
      break;
    if ( jitter )
    {
      if ( SleepTime * jitter / 0x64u )
        sub_18000E5F4();
    }
    SleepOnSleepMask();
  }
  free(v4);
  return sub_180019104();
}
```

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它通过判断 0x29 来决定是否直接调用 Sleep，如果使用了 sleep_mask，0x29 中存储的就是 text 段的结尾地址

```
void __fastcall sub_1800109F8(DWORD a1)
{
  if ( (unsigned int)GetIntValue(0x29u) && dword_180043430 <= 0 )
  {
    sub_180010B18();
    ((void (__fastcall *)(__int64, void (__stdcall *)(DWORD), _QWORD))lpAddress)(qword_18003E4C8, Sleep, a1);
  }
  else
  {
    Sleep(a1);
  }
}
```

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首先它会将自己当前函数与加解密函数一起传入

```
__int64 sub_180010B18()
{
  return sub_180010B2C(sub_180010B18, EncAndDec);
}
```

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在函数中先计算了两个函数地址的差值，通过对比很明显能够看出这个就是加解密函数的长度

```
void __fastcall sub_180010B2C(int a1, const void *a2)
{
  unsigned int v3; // edi
  __int64 v4; // rbx
  DWORD flOldProtect; // [rsp+40h] [rbp+18h] BYREF

  if ( !lpAddress )
  {
    v3 = a1 - (_DWORD)a2;
    v4 = (unsigned int)GetIntValue(0x29u);
    lpAddress = (LPVOID)(v4 + sub_1800185B8());
    VirtualProtect(lpAddress, v3, 4u, &flOldProtect);
    memmove(lpAddress, a2, (int)v3);
    VirtualProtect(lpAddress, v3, flOldProtect, &flOldProtect);
    qword_18003E4C8 = (__int64)malloc(0x20ui64);
    *(_QWORD *)qword_18003E4C8 = sub_1800185B8();
    *(_QWORD *)(qword_18003E4C8 + 8) = GetPtrValue(0x2Au);
    sub_180001218(qword_18003E4C8 + 16, 13i64, 0i64);
  }
}
```

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接着从 0x29 中取出 text 段的结尾地址，并从全局变量中取出 PE 头的位置，相加也就得到了在内存中的 text 的结尾地址，也就是后面要存放加解密函数的地址，接着传统的拷贝也就不提了，重点是后面的赋值操作

```
__int64 sub_1800185B8()
{
  return PEHeader;
}
```

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申请了 0x20 的空间，也就是 32 个字节，8 个字节存储 PE 头地址，8 个字节存储 0x2A 的地址

```
    memmove(lpAddress, a2, (int)v3);
    VirtualProtect(lpAddress, v3, flOldProtect, &flOldProtect);
    qword_18003E4C8 = (__int64)malloc(0x20ui64);
    *(_QWORD *)qword_18003E4C8 = sub_1800185B8();
    *(_QWORD *)(qword_18003E4C8 + 8) = GetPtrValue(0x2Au);
    sub_180001218(qword_18003E4C8 + 16, 13i64, 0i64);
  }
}
```

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这个 0x2A 也就是之前构造的那一串结构

```
Packer packer = new Packer();
packer.little();
if (!obfuscate) {
    packer.addInt( i: 0);
    packer.addInt( i: 4096);
}

Iterator sectionsTable = beaconParser.SectionsTable().iterator();

while(true) {
    String var10;
    do {
        if (!sectionsTable.hasNext()) {
            packer.addInt( i: 0);
            packer.addInt( i: 0);
            settings.addData(42, packer.getBytes(), (int)packer.size());
            return;
        }

        var10 = (String)sectionsTable.next();
    } while(".text".equals(var10) && !userwx);
```

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然后将加 16 的位置传入

```
__int64 __fastcall sub_180001218(BYTE *a1, DWORD a2, __int64 a3)
{
  __int64 result; // rax

  result = sub_180001184(a1, a2);
  if ( !(_DWORD)result )
    result = sub_1800010D0(a1, a2, a3);
  return result;
}
```

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第一个使用 CryptoAPI 生成密钥

```
DWORD __fastcall sub_180001184(BYTE *pbBuffer, DWORD dwLen)
{
  DWORD result; // eax
  HCRYPTPROV phProv; // [rsp+50h] [rbp+18h] BYREF

  phProv = 0i64;
  if ( CryptAcquireContextA(&phProv, 0i64, "Microsoft Base Cryptographic Provider v1.0", 1u, 0xF0000020)
    || (result = CryptAcquireContextA(&phProv, 0i64, "Microsoft Base Cryptographic Provider v1.0", 1u, 0xF0000028)) != 0 )
  {
    if ( !CryptGenRandom(phProv, dwLen, pbBuffer) )
      dwLen = 0;
    CryptReleaseContext(phProv, 0);
    result = dwLen;
  }
  return result;
}
```

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如果失败的话，用第二个自己实现的算法生成密钥

```
__int64 __fastcall sub_1800010D0(_BYTE *a1, unsigned int a2, void (*a3)(void))
{
  int v3; // er12
  int v4; // er14
  char v5; // bl
  unsigned int v7; // edi
  __int64 result; // rax
  int v10; // ebp
  clock_t v11; // esi
  clock_t v12; // esi

  v3 = 0;
  v4 = 0;
  v5 = 0;
  v7 = a2;
  result = a2;
  v10 = 8;
  if ( a2 )
  {
    do
    {
      --v7;
      if ( a3 )
        a3();
      do
      {
        --v10;
        do
        {
          v11 = clock();
          while ( v11 == clock() )
            v4 ^= 1u;
          v12 = clock();
          while ( v12 == clock() )
            v3 ^= 1u;
        }
        while ( v4 == v3 );
        v5 = v4 | (2 * v5);
      }
      while ( v10 );
      *a1 = v5;
      v5 = 0;
      ++a1;
      v10 = 8;
    }
    while ( v7 );
    result = a2;
```

```
  }
  return result;
}
```

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最后就来调用加解密函数了，第一个是构造的结构，第二个是 Sleep 函数，第三个是时间

```
void __fastcall sub_1800109F8(DWORD a1)
{
  if ( (unsigned int)GetIntValue(0x29u) && dword_180043430 <= 0 )
  {
    sub_180010B18();
    ((void (__fastcall *)(__int64, void (__stdcall *)(DWORD), _QWORD))lpAddress)(qword_18003E4C8, Sleep, a1);
  }
  else
  {
    Sleep(a1);
  }
}
```

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接着就来分析加解密函数，根据中间 Sleep 调用可以推断出，上面是加密函数，下面是解密函数

```
  v3 = (unsigned int *)a1[1];
  while ( 1 )
  {
    v6 = *v3;
    v7 = v3[1];
    v3 += 2;
    if ( !__PAIR64__(v7, v6) )
      break;
    if ( (unsigned int)v6 < v7 )
    {
      v8 = v6;
      do
      {
        v9 = v6;
        v10 = (unsigned int)v6 / 0xD;
        LODWORD(v6) = v6 + 1;
        *(_BYTE *)(*a1 + v8++) ^= *((_BYTE *)a1 + v9 - 13 * v10 + 16);
      }
      while ( (unsigned int)v6 < v7 );
    }
  }
  result = a2(a3);
  v12 = (unsigned int *)a1[1];
  while ( 1 )
  {
    v13 = *v12;
    v14 = v12[1];
    v12 += 2;
    if ( !__PAIR64__(v14, v13) )
      break;
    if ( (unsigned int)v13 < v14 )
    {
      v15 = v13;
      do
      {
        v16 = v13;
        v17 = (unsigned int)v13 / 0xD;
        LODWORD(v13) = v13 + 1;
        result = v16 - 13 * v17;
        *(_BYTE *)(*a1 + v15++) ^= *((_BYTE *)a1 + result + 16);
      }
      while ( (unsigned int)v13 < v14 );
    }
  }
  return result;
}
```

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参数是指针类型，取数组，下标 1，也就是 0x2A 所取到的值了

然后 v6 就是第一个值 sectionAddress，v7 就是第二个值 sectionEnd，然后将 v3 再后移

用来判断跳出规则的就是最后添加的两个零

do while 的混淆条件就是 sectionAddress<sectionEnd，也就是将所有的代码都进行混淆

```
v3 = (unsigned int *)a1[1];
while ( 1 )
{
  v6 = *v3;
  v7 = v3[1];
  v3 += 2;
  if ( !__PAIR64__(v7, v6) )
    break;
  if ( (unsigned int)v6 < v7 )
  {
    v8 = v6;
    do
    {
      v9 = v6;
      v10 = (unsigned int)v6 / 0xD;
      LODWORD(v6) = v6 + 1;
      *(_BYTE *)(*a1 + v8++) ^= *((_BYTE *)a1 + v9 - 13 * v10 + 16);
    }
    while ( (unsigned int)v6 < v7 );
  }
}
```

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之后的解密也是一样的逻辑就不提了

接下来就是验证的时候了

Sleep 时候的状态

```
0000000000190000    59                  pop rcx
0000000000190001    A8 BD               test al,BD
0000000000190003    F68C73 DA3B8685 44  test byte ptr ds:[rbx+rsi*2-7A79C426],4
000000000019000B    1D 0214F2B4         sbb eax,B4F21402
0000000000190010    29C4                sub esp,eax
0000000000190012    D1AC21 314C21E2     shr dword ptr ds:[rcx-1DDEB3CF],1
0000000000190019    4A:95               xchg rbp,rax
000000000019001B    3108                xor dword ptr ds:[rax],ecx
000000000019001D    FB                  sti
000000000019001E    D83B                fdivr st(0),dword ptr ds:[rbx]
0000000000190020    AC                  lodsb
0000000000190021    0D 8FBC5888         or eax,8858BC8F
0000000000190026    A0 429AF8A4D93B0996 mov al,byte ptr ds:[96093BD9A4F89A42]
000000000019002F    47:FD               std
0000000000190031    64:ED               in eax,dx
0000000000190033    0214F2              add dl,byte ptr ds:[rdx+rsi*8]
0000000000190036    FC                  cld
0000000000190037    A4                  movsb
0000000000190038    D93B                fnstcw word ptr ds:[rbx]
000000000019003A    53                  push rbx
000000000019003B    DE36                fidiv st(0),word ptr ds:[rsi]
000000000019003D    04 A8               add al,A8
000000000019003F    3D 0C0B48F2         cmp eax,F2480B0C
0000000000190044    A4                  movsb
0000000000190045    6D                  insd
0000000000190046    329E FF7605E4       xor bl,byte ptr ds:[rsi-1BFA8901]
000000000019004C    F0                  ???
000000000019004D    2340 9A             and eax,dword ptr ds:[rax-66]
0000000000190050    95                  xchg ebp,eax
0000000000190051    D7                  xlat
0000000000190052    F9                  stc
0000000000190053    4B:21B1 A976C950    and qword ptr ds:[r9+50C976A9],rsi
000000000019005A    2277 93             and dh,byte ptr ds:[rdi-6D]
000000000019005D    92                  xchg edx,eax
000000000019005E    CA B64F             ret far 4FB6
0000000000190061    73 BC               jae 19001F
0000000000190063    AB                  stosd
0000000000190064    24 DA               and al,DA
0000000000190066    48:6C               insb
0000000000190068    34 9B               xor al,9B
000000000019006A    92                  xchg edx,eax
000000000019006B    849D 7400FEA3       test byte ptr ss:[rbp-5C01FF8C],bl
0000000000190071    6BCC 58             imul ecx,esp,58
0000000000190074    2C 19               sub al,19
0000000000190076    FFF6                push rsi
0000000000190078    80D9 3B             sbb cl,3B
000000000019007B    53                  push rbx
000000000019007C    DECE                fmulp st(6),st(0)
000000000019007E    04 A8               add al,A8
0000000000190080    48                  ???
0000000000190081    D6                  ???
0000000000190082    C6                  ???
0000000000190083    9B                  fwait
0000000000190084    CD 11               int 11
0000000000190086    65:0162 6B          add dword ptr gs:[rdx+6B],esp
000000000019008A    72 3E               jb 1900CA
000000000019008C    99                  cdq
000000000019008D    88BE 2EA5A7D6       mov byte ptr ds:[rsi-29585AD2],bh
0000000000190093    E3 0B               jrcxz 1900A0
0000000000190095    E6 62               out 62,al
0000000000190097    F4                  hlt
0000000000190098    16                  ???
0000000000190099    F2:53               push rbx
000000000019009B    38BD 47409E76       cmp byte ptr ss:[rbp+769E4047],bh
00000000001900A1    2E:28E4             sub ah,ah
```

信安成长计划

此时的代码段

```
000000000001A0000   A5                  movsd
000000000001A0001   D93B                fnstcw word ptr ds:[rbx]
000000000001A0003   60                  ???
000000000001A0004   0C 86               or al,86
000000000001A0006   8F                  ???
000000000001A0007   60                  ???
000000000001A0008   79 8F               jns 19FF99
000000000001A000A   56                  push rsi
000000000001A000B   F0                  ???
000000000001A000C   14 37               adc al,37
000000000001A000E   60                  ???
000000000001A000F   3B53 96             cmp edx,dword ptr ds:[rbx-6A]
000000000001A0012   45:CB               ret far
000000000001A0014   40:C2 BB14          ret 14BB
000000000001A0018   F2:B9 97190881      mov ecx,81081997
000000000001A001E   96                  xchg esi,eax
000000000001A001F   45:CB               ret far
000000000001A0021   E4 B6               in al,B6
000000000001A0023   FA                  cli
000000000001A0024   FC                  cld
000000000001A0025   8945 A4             mov dword ptr ss:[rbp-5C],eax
000000000001A0028   D972 DE             fnstenv m28 ptr ds:[rdx-22]
000000000001A002B   91                  xchg ecx,eax
000000000001A002C   31BC56 C2FDEBBA     xor dword ptr ds:[rsi+rdx*2-4514023E],e
000000000001A0033   C7                  ???
000000000001A0034   6C                  insb
000000000001A0035   D6                  ???
000000000001A0036   BC 9DDECE04         mov esp,4CEDE9D
000000000001A003B   ED                  in eax,dx
000000000001A003C   0E                  ???
000000000001A003D   CB                  ret far
000000000001A003E   59                  pop rcx
000000000001A003F   79 38               jns 1A0079
000000000001A0041   E5 52               in eax,52
000000000001A0043   ED                  in eax,dx
000000000001A0044   1B55 00             sbb edx,dword ptr ss:[rbp]
000000000001A0047   FB                  sti
000000000001A0048   BD 9EC215F2         mov ebp,F215C29E
000000000001A004D   B8 2F29B61B         mov eax,1BB6292F
000000000001A0052   214F FD             and dword ptr ds:[rdi-3],ecx
000000000001A0055   57                  push rdi
000000000001A0056   3A02                cmp al,byte ptr ds:[rdx]
000000000001A0058   14 FD               adc al,FD
000000000001A005A   7B 07               jnp 1A0063
000000000001A005C   D93B                fnstcw word ptr ds:[rbx]
000000000001A005E   53                  push rbx
000000000001A005F   67                  ???
000000000001A0060   D6                  ???
000000000001A0061   04 A8               add al,A8
000000000001A0063   3D EABB6EFC         cmp eax,FC6EBBEA
000000000001A0068   A4                  movsb
000000000001A0069   91                  xchg ecx,eax
000000000001A006A   B0 5E               mov al,5E                          5E:'^'
000000000001A006C   E6 2A               out 2A,al
000000000001A006E   06                  ???
000000000001A006F   A8 79               test al,79
000000000001A0071   8911                mov dword ptr ds:[rcx],edx
000000000001A0073   CB                  ret far
000000000001A0074   18A6 D973DA96       sbb byte ptr ds:[rsi-69258C27],ah
000000000001A007A   DE40 21             fiadd st(0),word ptr ds:[rax+21]
000000000001A007D   3D 43EB32B4         cmp eax,B432EB43
000000000001A0082   2995 1F7364CE       sub dword ptr ss:[rbp-319B8CE1],edx
000000000001A0088   14 A8               adc al,A8
000000000001A008A   3D 4A9F2AB4         cmp eax,B42A9F4A
000000000001A008F   2D A1331757         sub eax,571733A1
```

信安成长计划

接收命令时候的状态

```
0000000000190000    4D:5A                pop r10                          r10:"Mozilla/5.0 (compatible; MSIE
0000000000190002    41:52                push r10                         r10:"Mozilla/5.0 (compatible; MSIE
0000000000190004    55                   push rbp
0000000000190005    48:89E5              mov rbp,rsp
0000000000190008    48:81EC 20000000     sub rsp,20
000000000019000F    48:8D1D EAFFFFFF     lea rbx,qword ptr ds:[190000]
0000000000190016    48:89DF              mov rdi,rbx
0000000000190019    48:81C3 F45F0100     add rbx,15FF4
0000000000190020    FFD3                 call rbx
0000000000190022    41:B8 F0B5A256       mov r8d,56A2B5F0
0000000000190028    68 04000000          push 4
000000000019002D    5A                   pop rdx
000000000019002E    48:89F9              mov rcx,rdi
0000000000190031    CC                   int3
0000000000190032    D000                 rol byte ptr ds:[rax],1
0000000000190034    0000                 add byte ptr ds:[rax],al
0000000000190036    0000                 add byte ptr ds:[rax],al
0000000000190038    0000                 add byte ptr ds:[rax],al
000000000019003A    0000                 add byte ptr ds:[rax],al
000000000019003C    F8                   clc
000000000019003D    0000                 add byte ptr ds:[rax],al
000000000019003F    000E                 add byte ptr ds:[rsi],cl
0000000000190041    1F                   ???
0000000000190042    BA 0E00B409          mov edx,9B4000E
0000000000190047    CD 21                int 21
0000000000190049    B8 014CCD21          mov eax,21CD4C01
000000000019004E    54                   push rsp
000000000019004F    68 69732070          push 70207369
0000000000190054    72 6F                jb 1900C5
0000000000190056    67:72 61             jb 1900BA
0000000000190059    6D                   insd
000000000019005A    2063 61              and byte ptr ds:[rbx+61],ah
000000000019005D    6E                   outsb
000000000019005E    6E                   outsb
000000000019005F    6F                   outsd
0000000000190060    74 20                je 190082
0000000000190062    62                   ???
0000000000190063    65:2072 75           and byte ptr gs:[rdx+75],dh
0000000000190067    6E                   outsb
0000000000190068    2069 6E              and byte ptr ds:[rcx+6E],ch
000000000019006B    20444F 53            and byte ptr ds:[rdi+rcx*2+53],al
000000000019006F    206D 6F              and byte ptr ss:[rbp+6F],ch
0000000000190072    64652E:0D 0D0A2400   or eax,240A0D
000000000019007A    0000                 add byte ptr ds:[rax],al
000000000019007C    0000                 add byte ptr ds:[rax],al
000000000019007E    0000                 add byte ptr ds:[rax],al
0000000000190080    75 D4                jne 190056
0000000000190082    D269 31              shr byte ptr ds:[rcx+31],cl
0000000000190085    B5 BC                mov ch,BC
0000000000190087    3A31                 cmp dh,byte ptr ds:[rcx]
0000000000190089    B5 BC                mov ch,BC
000000000019008B    3A31                 cmp dh,byte ptr ds:[rcx]
000000000019008D    B5 BC                mov ch,BC
000000000019008F    3A57 5B              cmp dl,byte ptr ds:[rdi+5B]
0000000000190092    72 3A                jb 1900CE
0000000000190094    30B5 BC3A125A        xor byte ptr ss:[rbp+5A123ABC],dh
000000000019009A    6E                   outsb
000000000019009B    3AA9 B5BC3AAF        cmp ch,byte ptr ds:[rcx-50C5434B]
00000000001900A1    15 7B3A30B5          adc eax,B5303A7B
00000000001900A6    BC 3AC07373          mov esp,7373C03A
00000000001900AB    3A18                 cmp bl,byte ptr ds:[rax]
00000000001900AD    B5 BC                mov ch,BC
00000000001900AF    3AC0                 cmp al,al
00000000001900B1    73 72                jae 190125
00000000001900B3    3AB8 B5BC3AC0        cmp bh,byte ptr ds:[rax-3FC5434B]
```

信安成长计划

此时的代码段

```
000000000001A0000   0100          add dword ptr ds:[rax],eax
000000000001A0002   0033          add byte ptr ds:[rbx],dh
000000000001A0004   D248 8B       ror byte ptr ds:[rax-75],cl
000000000001A0007   C8 448D 42    enter 8D44,42
000000000001A000B   02E8          add ch,al
000000000001A000D   93            xchg ebx,eax
000000000001A000E   B9 0000488B   mov ecx,8B480000
000000000001A0013   CF            iretd
000000000001A0014   E8 FFB90000   call 1ABA18
000000000001A0019   45:33C0       xor r8d,r8d
000000000001A001C   33D2          xor edx,edx
000000000001A001E   48:8BCF       mov rcx,rdi
000000000001A0021   4C:8BF8       mov r15,rax
000000000001A0024   E8 7BB90000   call 1AB9A4
000000000001A0029   49:8D4F FF    lea rcx,qword ptr ds:[r15-1]
000000000001A002D   B8 FEFFFFFF   mov eax,FFFFFFFE
000000000001A0032   48:3BC8       cmp rcx,rax
000000000001A0035   0F87 CE000000 ja 1A0109
000000000001A003B   45:33C9       xor r9d,r9d
000000000001A003E   4D:8BC4       mov r8,r12
000000000001A0041   41:8BD6       mov edx,r14d
000000000001A0044   48:8BCE       mov rcx,rsi
000000000001A0047   FF15 A3C00100 call qword ptr ds:[<&GetFullPathNameA>]
000000000001A004D   44:8BF0       mov r14d,eax
000000000001A0050   8D48 FF       lea ecx,qword ptr ds:[rax-1]
000000000001A0053   81F9 FF070000 cmp ecx,7FF
000000000001A0059   0F87 A3000000 ja 1A0102
000000000001A005F   B9 18000000   mov ecx,18
000000000001A0064   E8 AF9C0000   call 1A9D18
000000000001A0069   48:8B0D 38E40200 mov rcx,qword ptr ds:[1CE4A8]
000000000001A0070   44:8B05 39E40200 mov r8d,dword ptr ds:[1CE4B0]
000000000001A0077   48:8948 10    mov qword ptr ds:[rax+10],rcx
000000000001A007B   44:8900       mov dword ptr ds:[rax],r8d
000000000001A007E   41:FFC0       inc r8d
000000000001A0081   48:8D4C24 20  lea rcx,qword ptr ss:[rsp+20]     [rsp+20]:L"ｔ繼鯛"
000000000001A0086   BA 00100000   mov edx,1000
000000000001A008B   48:8BD8       mov rbx,rax
000000000001A008E   48:8978 08    mov qword ptr ds:[rax+8],rdi
000000000001A0092   44:8978 04    mov dword ptr ds:[rax+4],r15d
000000000001A0096   48:8905 0BE40200 mov qword ptr ds:[1CE4A8],rax
000000000001A009D   44:8905 0CE40200 mov dword ptr ds:[1CE4B0],r8d
000000000001A00A4   E8 8BCEFFFF   call 19CF34
000000000001A00A9   8B13          mov edx,dword ptr ds:[rbx]
000000000001A00AB   48:8D4C24 20  lea rcx,qword ptr ss:[rsp+20]     [rsp+20]:L"ｔ繼鯛"
000000000001A00B0   E8 AFCEFFFF   call 19CF64
000000000001A00B5   48:8D4C24 20  lea rcx,qword ptr ss:[rsp+20]     [rsp+20]:L"ｔ繼鯛"
000000000001A00BA   41:8BD7       mov edx,r15d
000000000001A00BD   E8 A2CEFFFF   call 19CF64
000000000001A00C2   48:8D4C24 20  lea rcx,qword ptr ss:[rsp+20]     [rsp+20]:L"ｔ繼鯛"
000000000001A00C7   45:8BC6       mov r8d,r14d
000000000001A00CA   49:8BD4       mov rdx,r12
000000000001A00CD   E8 D6CDFFFF   call 19CEA8
000000000001A00D2   48:8D4C24 20  lea rcx,qword ptr ss:[rsp+20]     [rsp+20]:L"ｔ繼鯛"
000000000001A00D7   E8 983D0000   call 1A3E74
000000000001A00DC   48:8D4C24 20  lea rcx,qword ptr ss:[rsp+20]     [rsp+20]:L"ｔ繼鯛"
000000000001A00E1   8BD8          mov ebx,eax
000000000001A00E3   E8 18CFFFFF   call 19D000
000000000001A00E8   41:B8 02000000 mov r8d,2
000000000001A00EE   48:8BC8       mov rcx,rax
000000000001A00F1   8BD3          mov edx,ebx
000000000001A00F3   41:FFD5       call r13
000000000001A00F6   48:8D4C24 20  lea rcx,qword ptr ss:[rsp+20]     [rsp+20]:L"ｔ繼鯛"
000000000001A00FB   E8 10CEFFFF   call 19CF10
000000000001A0100   EB 33         jmp 1A0135
000000000001A0102   B9 3D000000   mov ecx,3D                        3D:'='
```

信安成长计划

在允许 RWX 权限的时候，代码段是混淆的

# 0x03 set userwx "false"

理论在之前也都讲完了，这里就是验证一下最终的效果

Sleep 时候的状态

```
0000000000A20000   6F                 outsd
0000000000A20001   64                 ???
0000000000A20002   17                 ???
0000000000A20003   FC                 cld
0000000000A20004   4F:ED              in eax,dx
0000000000A20006   66:0B59 58         or bx,word ptr ds:[rcx+58]
0000000000A2000A   E4 B2              in al,B2
0000000000A2000C   26:223E            and bh,byte ptr es:[rsi]
0000000000A2000F   1E                 ???
0000000000A20010   2307               and eax,dword ptr ds:[rdi]
0000000000A20012   4F:1011            adc byte ptr ds:[r9],r10b
0000000000A20015   EE                 out dx,al
0000000000A20016   91                 xchg ecx,eax
0000000000A20017   814D 6E A3FDA2F1   or dword ptr ss:[rbp+6E],F1A2FDA3
0000000000A2001E   1BA5 103D5061      sbb esp,dword ptr ss:[rbp+61503D10]
0000000000A20024   F8                 clc
0000000000A20025   27                 ???
0000000000A20026   847456 52          test byte ptr ds:[rsi+rdx^2+52],dh
0000000000A2002A   AE                 scasb
0000000000A2002B   1AA5 B5A69820      sbb ah,byte ptr ss:[rbp+2098A6B5]
0000000000A20031   C4                 ???
0000000000A20032   4226:223E          and bh,byte ptr es:[rsi]
0000000000A20036   56                 push rsi
0000000000A20037   AE                 scasb
0000000000A20038   1AA5 EFEEE9D9      sbb ah,byte ptr ss:[rbp-26161111]
0000000000A2003E   0892 283D8458      or byte ptr ds:[rdx+58843D28],dl
0000000000A20044   AE                 scasb
0000000000A20045   AE                 scasb
0000000000A20046   AC                 lodsb
0000000000A20047   22CF               and cl,bh
0000000000A20049   A9 D8445F07        test eax,75F44D8
0000000000A2004E   76 56              jbe A200A6
0000000000A20050   3F                 ???
0000000000A20051   DD3A               fnstsw word ptr ds:[rdx]
0000000000A20053   D5                 ???
0000000000A20054   9D                 popfq
0000000000A20055   8176 AB 69FF0641   xor dword ptr ds:[rsi-55],4106FF69
0000000000A2005C   5F                 pop rdi
0000000000A2005D   38C0               cmp al,al
0000000000A2005F   75 D1              jne A20032
0000000000A20061   CF                 iretd
0000000000A20062   8C                 ???
0000000000A20063   74 F9              je A2005E
0000000000A20065   7A E7              jp A2004E
0000000000A20067   48:0257 38         add dl,byte ptr ds:[rdi+38]
0000000000A2006B   8E5E EA            mov ds,word ptr ds:[rsi-16]
0000000000A2006E   BC CE7CB66C        mov esp,6CB67CCE
0000000000A20073   F708 2F335C8A      test dword ptr ds:[rax],8A5C332F
0000000000A20079   1AA5 EFEE11D9      sbb ah,byte ptr ss:[rbp-26EE1111]
0000000000A2007F   08E7               or bh,ah
0000000000A20081   F2                 ???
0000000000A20082   F0                 ???
0000000000A20083   57                 push rdi
0000000000A20084   67:1BA6 9FDE5BAD   sbb esp,dword ptr ds:[esi-52A42161]
0000000000A2008B   E3 39              jrcxz A200C6
0000000000A2008D   27                 ???
0000000000A2008E   9A                 ???
0000000000A2008F   1869 0D            sbb byte ptr ds:[rcx+D],ch
0000000000A20092   DC20               fsub st(0),qword ptr ds:[rax]
0000000000A20094   95                 xchg ebp,eax
0000000000A20095   5A                 pop rdx
0000000000A20096   52                 push rdx
0000000000A20097   2BCB               sub ecx,ebx
0000000000A20099   52                 push rdx
0000000000A2009A   FC                 cld
```

信安成长计划

此时的代码段

```
0000000000A2FFFD   0F87 25010000     ja A30128
0000000000A30003   33D2              xor edx,edx
0000000000A30005   48:8BC8           mov rcx,rax
0000000000A30008   44:8D42 02        lea r8d,qword ptr ds:[rdx+2]
0000000000A3000C   E8 93B90000       call A3B9A4
0000000000A30011   48:8BCF           mov rcx,rdi
0000000000A30014   E8 FFB90000       call A3BA18
0000000000A30019   45:33C0           xor r8d,r8d
0000000000A3001C   33D2              xor edx,edx
0000000000A3001E   48:8BCF           mov rcx,rdi
0000000000A30021   4C:8BF8           mov r15,rax
0000000000A30024   E8 7BB90000       call A3B9A4
0000000000A30029   49:8D4F FF        lea rcx,qword ptr ds:[r15-1]
0000000000A3002D   B8 FEFFFFFF       mov eax,FFFFFFFE
0000000000A30032   48:3BC8           cmp rcx,rax
0000000000A30035   0F87 CE000000     ja A30109
0000000000A3003B   45:33C9           xor r9d,r9d
0000000000A3003E   4D:8BC4           mov r8,r12
0000000000A30041   41:8BD6           mov edx,r14d
0000000000A30044   48:8BCE           mov rcx,rsi
0000000000A30047   FF15 A3C00100     call qword ptr ds:[A4C0F0]
0000000000A3004D   44:8BF0           mov r14d,eax
0000000000A30050   8D48 FF           lea ecx,qword ptr ds:[rax-1]
0000000000A30053   81F9 FF070000     cmp ecx,7FF
0000000000A30059   0F87 A3000000     ja A30102
0000000000A3005F   B9 18000000       mov ecx,18
0000000000A30064   E8 AF9C0000       call A39D18
0000000000A30069   48:8B0D 38E40200  mov rcx,qword ptr ds:[A5E4A8]
0000000000A30070   44:8B05 39E40200  mov r8d,dword ptr ds:[A5E4B0]
0000000000A30077   48:8948 10        mov qword ptr ds:[rax+10],rcx
0000000000A3007B   44:8900           mov dword ptr ds:[rax],r8d
0000000000A3007E   41:FFC0           inc r8d
0000000000A30081   48:8D4C24 20      lea rcx,qword ptr ss:[rsp+20]
0000000000A30086   BA 00100000       mov edx,1000
0000000000A3008B   48:8BD8           mov rbx,rax
0000000000A3008E   48:8978 08        mov qword ptr ds:[rax+8],rdi
0000000000A30092   44:8978 04        mov dword ptr ds:[rax+4],r15d
0000000000A30096   48:8905 0BE40200  mov qword ptr ds:[A5E4A8],rax
0000000000A3009D   44:8905 0CE40200  mov dword ptr ds:[A5E4B0],r8d
0000000000A300A4   E8 8BCEFFFF       call A2CF34
0000000000A300A9   8B13              mov edx,dword ptr ds:[rbx]
0000000000A300AB   48:8D4C24 20      lea rcx,qword ptr ss:[rsp+20]
0000000000A300B0   E8 AFCEFFFF       call A2CF64
0000000000A300B5   48:8D4C24 20      lea rcx,qword ptr ss:[rsp+20]
0000000000A300BA   41:8BD7           mov edx,r15d
0000000000A300BD   E8 A2CEFFFF       call A2CF64
0000000000A300C2   48:8D4C24 20      lea rcx,qword ptr ss:[rsp+20]
0000000000A300C7   45:8BC6           mov r8d,r14d
0000000000A300CA   49:8BD4           mov rdx,r12
0000000000A300CD   E8 D6CDFFFF       call A2CEA8
0000000000A300D2   48:8D4C24 20      lea rcx,qword ptr ss:[rsp+20]
0000000000A300D7   E8 983D0000       call A33E74
0000000000A300DC   48:8D4C24 20      lea rcx,qword ptr ss:[rsp+20]
0000000000A300E1   8BD8              mov ebx,eax
0000000000A300E3   E8 18CFFFFF       call A2D000
0000000000A300E8   41:B8 02000000    mov r8d,2
0000000000A300EE   48:8BC8           mov rcx,rax
0000000000A300F1   8BD3              mov edx,ebx
0000000000A300F3   41:FFD5           call r13
0000000000A300F6   48:8D4C24 20      lea rcx,qword ptr ss:[rsp+20]
0000000000A300FB   E8 10CEFFFF       call A2CF10
0000000000A30100   EB 33             jmp A30135
0000000000A30102   B9 3D000000       mov ecx,3D                    3D:'='
0000000000A30107   EB 05             jmp A3010E
0000000000A30109   B9 3C000000       mov ecx,3C                    3C:'<'
```

信安成长计划

接收命令时的状态

```
0000000000A20000   4D:5A                 pop r10
0000000000A20002   41:52                 push r10
0000000000A20004   55                    push rbp
0000000000A20005   48:89E5               mov rbp,rsp
0000000000A20008   48:81EC 20000000      sub rsp,20
0000000000A2000F   48:8D1D EAFFFFFF      lea rbx,qword ptr ds:[A20000]
0000000000A20016   48:89DF               mov rdi,rbx
0000000000A20019   48:81C3 F45F0100      add rbx,15FF4
0000000000A20020   FFD3                  call rbx
0000000000A20022   41:B8 F0B5A256        mov r8d,56A2B5F0
0000000000A20028   68 04000000           push 4
0000000000A2002D   5A                    pop rdx
0000000000A2002E   48:89F9               mov rcx,rdi
0000000000A20031   CC                    int3
0000000000A20032   D000                  rol byte ptr ds:[rax],1
0000000000A20034   0000                  add byte ptr ds:[rax],al
0000000000A20036   0000                  add byte ptr ds:[rax],al
0000000000A20038   0000                  add byte ptr ds:[rax],al
0000000000A2003A   0000                  add byte ptr ds:[rax],al
0000000000A2003C   F8                    clc
0000000000A2003D   0000                  add byte ptr ds:[rax],al
0000000000A2003F   000E                  add byte ptr ds:[rsi],cl
0000000000A20041   1F                    ???
0000000000A20042   BA 0E00B409           mov edx,9B4000E
0000000000A20047   CD 21                 int 21
0000000000A20049   B8 014CCD21           mov eax,21CD4C01
0000000000A2004E   54                    push rsp
0000000000A2004F   68 69732070           push 70207369
0000000000A20054   72 6F                 jb A200C5
0000000000A20056   67:72 61              jb A200BA
0000000000A20059   6D                    insd
0000000000A2005A   2063 61               and byte ptr ds:[rbx+61],ah
0000000000A2005D   6E                    outsb
0000000000A2005E   6E                    outsb
0000000000A2005F   6F                    outsd
0000000000A20060   74 20                 je A20082
0000000000A20062   62                    ???
0000000000A20063   65:2072 75            and byte ptr gs:[rdx+75],dh
0000000000A20067   6E                    outsb
0000000000A20068   2069 6E               and byte ptr ds:[rcx+6E],ch
0000000000A2006B   20444F 53             and byte ptr ds:[rdi+rcx*2+53],al
0000000000A2006F   206D 6F               and byte ptr ss:[rbp+6F],ch
0000000000A20072   64652E:0D 0D0A2400    or eax,240A0D
0000000000A2007A   0000                  add byte ptr ds:[rax],al
0000000000A2007C   0000                  add byte ptr ds:[rax],al
0000000000A2007E   0000                  add byte ptr ds:[rax],al
0000000000A20080   75 D4                 jne A20056
0000000000A20082   D269 31               shr byte ptr ds:[rcx+31],cl
0000000000A20085   B5 BC                 mov ch,BC
0000000000A20087   3A31                  cmp dh,byte ptr ds:[rcx]
0000000000A20089   B5 BC                 mov ch,BC
0000000000A2008B   3A31                  cmp dh,byte ptr ds:[rcx]
0000000000A2008D   B5 BC                 mov ch,BC
0000000000A2008F   3A57 5B               cmp dl,byte ptr ds:[rdi+5B]
0000000000A20092   72 3A                 jb A200CE
0000000000A20094   30B5 BC3A125A         xor byte ptr ss:[rbp+5A123ABC],dh
0000000000A2009A   6E                    outsb
0000000000A2009B   3AA9 B5BC3AAF         cmp ch,byte ptr ds:[rcx-50C5434B]
0000000000A200A1   15 7B3A30B5           adc eax,B5303A7B
0000000000A200A6   BC 3AC07373           mov esp,7373C03A
0000000000A200AB   3A18                  cmp bl,byte ptr ds:[rax]
0000000000A200AD   B5 BC                 mov ch,BC
0000000000A200AF   3AC0                  cmp al,al
0000000000A200B1   73 72                 jae A20125
0000000000A200B3   3AB8 B5BC3AC0         cmp bh,byte ptr ds:[rax-3FC5434B]
```

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此时的代码段

```
0000000000A2FFFD  0F87 25010000     ja A30128
0000000000A30003  33D2              xor edx,edx
0000000000A30005  48:8BC8           mov rcx,rax
0000000000A30008  44:8D42 02        lea r8d,qword ptr ds:[rdx+2]
0000000000A3000C  E8 93B90000       call A3B9A4
0000000000A30011  48:8BCF           mov rcx,rdi
0000000000A30014  E8 FFB90000       call A3BA18
0000000000A30019  45:33C0           xor r8d,r8d
0000000000A3001C  33D2              xor edx,edx
0000000000A3001E  48:8BCF           mov rcx,rdi
0000000000A30021  4C:8BF8           mov r15,rax
0000000000A30024  E8 7BB90000       call A3B9A4
0000000000A30029  49:8D4F FF        lea rcx,qword ptr ds:[r15-1]
0000000000A3002D  B8 FEFFFFFF       mov eax,FFFFFFFE
0000000000A30032  48:3BC8           cmp rcx,rax
0000000000A30035  0F87 CE000000     ja A30109
0000000000A3003B  45:33C9           xor r9d,r9d
0000000000A3003E  4D:8BC4           mov r8,r12
0000000000A30041  41:8BD6           mov edx,r14d
0000000000A30044  48:8BCE           mov rcx,rsi
0000000000A30047  FF15 A3C00100     call qword ptr ds:[<&GetFullPathNameA>]
0000000000A3004D  44:8BF0           mov r14d,eax
0000000000A30050  8D48 FF           lea ecx,qword ptr ds:[rax-1]
0000000000A30053  81F9 FF070000     cmp ecx,7FF
0000000000A30059  0F87 A3000000     ja A30102
0000000000A3005F  B9 18000000       mov ecx,18
0000000000A30064  E8 AF9C0000       call A39D18
0000000000A30069  48:8B0D 38E40200  mov rcx,qword ptr ds:[A5E4A8]
0000000000A30070  44:8B05 39E40200  mov r8d,dword ptr ds:[A5E4B0]
0000000000A30077  48:8948 10        mov qword ptr ds:[rax+10],rcx
0000000000A3007B  44:8900           mov dword ptr ds:[rax],r8d
0000000000A3007E  41:FFC0           inc r8d
0000000000A30081  48:8D4C24 20      lea rcx,qword ptr ss:[rsp+20]
0000000000A30086  BA 00100000       mov edx,1000
0000000000A3008B  48:8BD8           mov rbx,rax
0000000000A3008E  48:8978 08        mov qword ptr ds:[rax+8],rdi
0000000000A30092  44:8978 04        mov dword ptr ds:[rax+4],r15d
0000000000A30096  48:8905 0BE40200  mov qword ptr ds:[A5E4A8],rax
0000000000A3009D  44:8905 0CE40200  mov dword ptr ds:[A5E4B0],r8d
0000000000A300A4  E8 8BCEFFFF       call A2CF34
0000000000A300A9  8B13              mov edx,dword ptr ds:[rbx]
0000000000A300AB  48:8D4C24 20      lea rcx,qword ptr ss:[rsp+20]
0000000000A300B0  E8 AFCEFFFF       call A2CF64
0000000000A300B5  48:8D4C24 20      lea rcx,qword ptr ss:[rsp+20]
0000000000A300BA  41:8BD7           mov edx,r15d
0000000000A300BD  E8 A2CEFFFF       call A2CF64
0000000000A300C2  48:8D4C24 20      lea rcx,qword ptr ss:[rsp+20]
0000000000A300C7  45:8BC6           mov r8d,r14d
0000000000A300CA  49:8BD4           mov rdx,r12
0000000000A300CD  E8 D6CDFFFF       call A2CEA8
0000000000A300D2  48:8D4C24 20      lea rcx,qword ptr ss:[rsp+20]
0000000000A300D7  E8 983D0000       call A33E74
0000000000A300DC  48:8D4C24 20      lea rcx,qword ptr ss:[rsp+20]
0000000000A300E1  8BD8              mov ebx,eax
0000000000A300E3  E8 18CFFFFF       call A2D000
0000000000A300E8  41:B8 02000000    mov r8d,2
0000000000A300EE  48:8BC8           mov rcx,rax
0000000000A300F1  8BD3              mov edx,ebx
0000000000A300F3  41:FFD5           call r13
0000000000A300F6  48:8D4C24 20      lea rcx,qword ptr ss:[rsp+20]
0000000000A300FB  E8 10CEFFFF       call A2CF10
0000000000A30100  EB 33             jmp A30135
0000000000A30102  B9 3D000000       mov ecx,3D                    3D:'='
0000000000A30107  EB 05             jmp A3010E
0000000000A30109  B9 3C000000       mov ecx,3C                    3C:'<'
```

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在禁止 RWX 权限的时候，代码段是不混淆的

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